

Related Work

This paper builds on the recent work of [?] that introduces the idea of the compatibility score of an update (Equation 1) and proposes a method for increasing this score by employing a customized loss function (Equation 2) where an additional weighted penalty is given for newly introduced errors (mistakes that the model prior to the update didn't make). They showed that forcing the update to be more compatible generally decreases its performance, i.e. a performance-compatibility tradeoff. We expand this method by adding the notion of personalization towards target users with the goal of improving the performance-compatibility tradeoff provided by the update.

The underlying idea behind the method proposed by [?] (and therefore behind the method proposed here as well) is similar to several other works. One such example are Model Ensemble methods [?], in particular AdaBoost [?]. In both methods, an additional penalty is given for different types of errors that depends on a previously trained model. In AdaBoost this additional penalty is given for mistakes that the previous model made and in the method of [?] for mistakes that the previous model didn't make. It could be interesting to explore the theoretical similarities between these two methods, since model ensemble enjoys a vast theoretical framework [?].

Choosing the best model for each user is related to research on methods for choosing the best expert [?], but in our work we simply consider the quality of the performance-compatibility tradeoffs (in terms of AUC) provided by the various models on a validation set to determine this. Further implementation of the ideas proposed in that research may improve the reliability of this selection.

Several other works relate to the personalization of AI-models to users but do not address the personalization of updates to these systems, let alone the notion of an update's compatibility with the prior model. For instance, for the ASSISTment dataset mentioned in previous sections, work was performed on individualizing student models [?] and on clustering the students [?] with the goal of improving the accuracy of the predictions.

Much work has been done in the field of human-AI interactions. The compatibility of an update to an AI-system is closely related to the 14th Guideline for Human-AI Interactions from Amershi et al.'s work [?] described as "*Update and adapt cautiously: Limit disruptive changes when updating and adapting the AI-system's behaviors*". In our case, this means making sure that the predictions made by the updated model conform to the user's expectations that developed prior to the update. It is related also to the 5th step in an article from Google Design [?] that states the importance of making sure that the AI-system and the user's model evolve in tandem. For more references to related work on human-AI interaction and the field of AI-advised human decision making refer to the related work section in the paper of [?].

Conclusion

The compatibility of an update to an AI-system with the system prior to the update is important for the adequate func-

tioning of human-AI teams [?]. Previous work addressed the problem of increasing compatibility by developing a loss function that delivers an additional penalty for newly introduced errors (mistakes that the model prior to the update didn't make), and showed that there's a tradeoff between the compatibility and performance of the updated model [?]. We extended this approach by personalizing the model's objective function to target users with the goal of producing improving this tradeoff. We also proposed a framework for selecting the best way of performing this personalization.

The experimental results showed that our personalization approach can yield significantly better performance-compatibility tradeoffs than the baseline non-personalized model. We then analyzed two use cases where the personalization exceptionally outperformed the baseline and showed that the personalized classifier model differed fundamentally from the baseline model.

Our approach is limited in the sense that it assumes that the user's future interactions with the system will resemble the ones observed so far. In future work we will address this limitation, and explore ways of modifying the objective function beyond simply assigning weights to the dataset samples possibly by employing program synthesis or inverse reinforcement learning methods. We believe that informing users about the performance-compatibility tradeoff of the models that are used to interact with them can contribute on making AI-systems more transparent.

Acknowledgements

Thanks very much to Avi Segal and Nicholas Hoernle for helpful comments. This research was partially supported by Israeli Science Foundation (ISF) Grant No. 773/16 and by Canada's CIFAR AI Chairs program.

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